

Team Member Contribution

Roles, ownership, deliverables & contribution share for each team member.

Md Mohaiminul Hoq Alvi		ID 23101158
HARDWARE LEAD	Contribution share: 33.33%	
HARDWARE	• Mechanical assembly of 4WD chassis • Sensor & motor wiring (IR x3) • Soldering & cable management • Mounting servo, pump and water-spray nozzle	
SOFTWARE / FIRMWARE	• Pin-mapping review with firmware team	
TESTING	• Power-rail testing (5V regulation) • Manual drive tests of the rover platform	
DOCUMENTATION	• Hardware section of the main report	

Sultan Ruhul Kuddus Nayan		ID 23101159
FIRMWARE LEAD	Contribution share: 33.33%	
HARDWARE	• Cirkits Designer simulation of full circuit	
SOFTWARE / FIRMWARE	• Arduino C/C++ firmware development • IR sensor fusion AND-logic • L298N motor drive + steering routine • Servo aim + relay-controlled pump logic	
TESTING	• Unit tests for each sensor + actuator • Calibrating IR & temperature thresholds	
DOCUMENTATION	• Methodology section of report	

COMMS & DOCUMENTATION	Contribution share: 33.33%
HARDWARE	• Component procurement & inventory
SOFTWARE / FIRMWARE	• ESP8266 SMTP email-alert code • NEO-6M GPS NMEA parsing (TinyGPS++) • Google Maps link formatting + hotline injection
TESTING	• End-to-end integration tests • Performance benchmarking (detection latency, false-alarm rate, alert delivery)
DOCUMENTATION	• Full report compilation & formatting • Presentation slides + annexure • References & IEEE-style citations

Combined Contribution Summary

Phase / Deliverable	Alvi (23101158)	Nayan (23101159)	Lemon (23101160)
Hardware build & wiring	Lead	Support	Procure
Firmware & sensor fusion	Review	Lead	Support
GPS + Wi-Fi + Email alert	Support	Support	Lead
Testing & benchmarking	Lead	Lead	Lead
Report writing	Support	Support	Lead
Presentation & annexure	Support	Support	Lead
Demo & defense	Equal	Equal	Equal